# Ninel **BĂNICĂ**

*Embedded & Software Engineer*

## CONTACT & INFO


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/Ninel-Valentin

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## LANGUAGES

| | |
|---|---|
| Romanian | **Fluent** |
| English | **C1** |
| French | **B1** |

## SOFT SKILLS

- Problem-Solving
- Quick learner
- Curiosity
- Communication
- Time Management

## TECHNICAL SKILLS

Beginner - Expert

| | |
|---|---|
| Embedded | ●●● |
| C/C++ | ●●●● |
| C# | ●●● |
| Java | ●●● |
| Python | ●● |
| JavaScript | ●●●●● |
| HTML&CSS | ●●●●● |
| NodeJS | ●●●●● |
| ReactJS | ●●●● |
| AWS | ●● |
| Unity | ●● |

## Education

### Bachelor of Electrical Engineering & Computer Science

*Transylvania University of Brasov* **2022 - 2026**

## Experience

### FPGA usage and development - Practical work

*Univeristy Transylvania of Brasov* **Jun 2025 - Sep 2025**

- Learnt using Vivado and Vitis to program FPGAs and processors on it.
- Researched and learnt about SEU effects on hardware, error correction and error pattern observation and prevention

### Digital Twin developer intern

*Siemens Digital Industries Software* **Jun 2025 - Present**

- Improved an existing digital twin excavator arm electrical connections
- Changed the software for the control and simulation to work in real time.

### Full stack developer intern

*Siemens Digital Industries Software* **Nov 2024 - Jun 2025**

- Worked on a C# and C++ desktop app for data analysis of automotive sensors;

### Summer intern

*Siemens Digital Industries Software* **Jul 2024 - Sep 2024**

- Developed multiple microservices using Python and deployed them online using AWS

### Junior Crawler Developer

*ChannelSight* **Mar 2022 - Nov 2024**

- Developed and improved Web-Crawlers on Apify using NodeJs.
- Wrote custom JavaScript tools to ensure data quality in large volumes.
- Created SQL queries for data preview.

## Certifications/Courses

- AWS Cloud Practitioner **Oct 2024**
- SQL Server fundamentals - PluralSight **Jul 2023**

## Relevant personal projects

### Embedded system for satellite imagery and sensor data transmission

**2025**

- Designed a multi-sensor, dual-cam, ESP32 controlled PCB with an integrated antenna to emulate a satellite.
- Acquire sensor data and satellite imagery to send to a ground receiver

### Gyrosensor controlled AVC recovering device

*Arduino (**C/C++**)* **2025**

- Built a glove with a gyrosensor and bluetooth module that sends control commands to a bluetooth LCD application that helps dexterity.

### Graph Pathfinding Visualizer

*Java* **2024**

- Interactive visual canvas used for editing directed/undirected graphs.
- Find the shortest path between a starting and ending point.

*Many more projects that I've worked on since 2021 available on my portfolio website.*